Homework 03

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Set up

```
# read in packages
library(tidyverse)
library(here)
library(janitor)
library(readxl)
library(ggeffects)

# store personal data
my_data <- read_csv(here("data" , "personaldataproject.csv"))

# store kelp data
temp_kelp <- read_csv(here("data", "temp-kelp.csv"))
```

Problems

Problem 1. Giant Kelp Fronds

a. An Appropriate Test

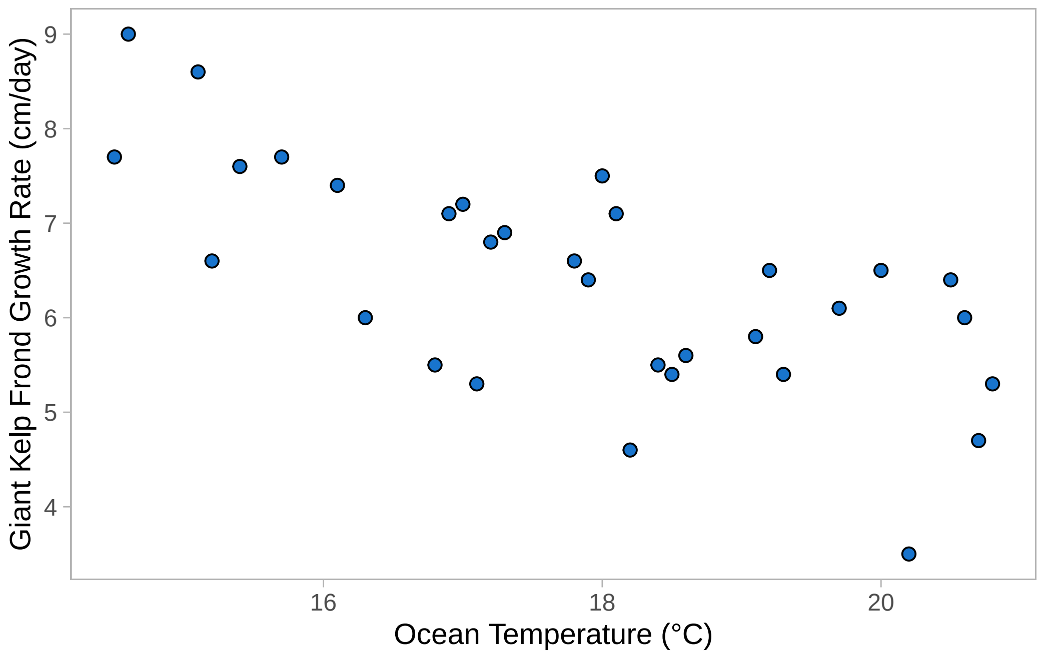
Pearson and Spearman correlations could be used to determine strength of the relationship between ocean temperature and giant kelp frond elongation rate assuming observations are independent. Pearson's correlation also assumes a linear relationship exists and it is a parametric test meaning it assumes variables are normally distributed. Spearman's correlation assumes a monotonic relationship and it is a nonparametric test meaning normality is not an assumption.

b. Visualizing Data

Showing the relationship between ocean temperature and giant kelp frond elongation rate.

```
# plot raw data points to observe potential linear relationship

# base layer ggplot
ggplot(data = temp_kelp,
       # assign data to axes
       aes(x = temp_c,
           y = kelp_elong))+
```

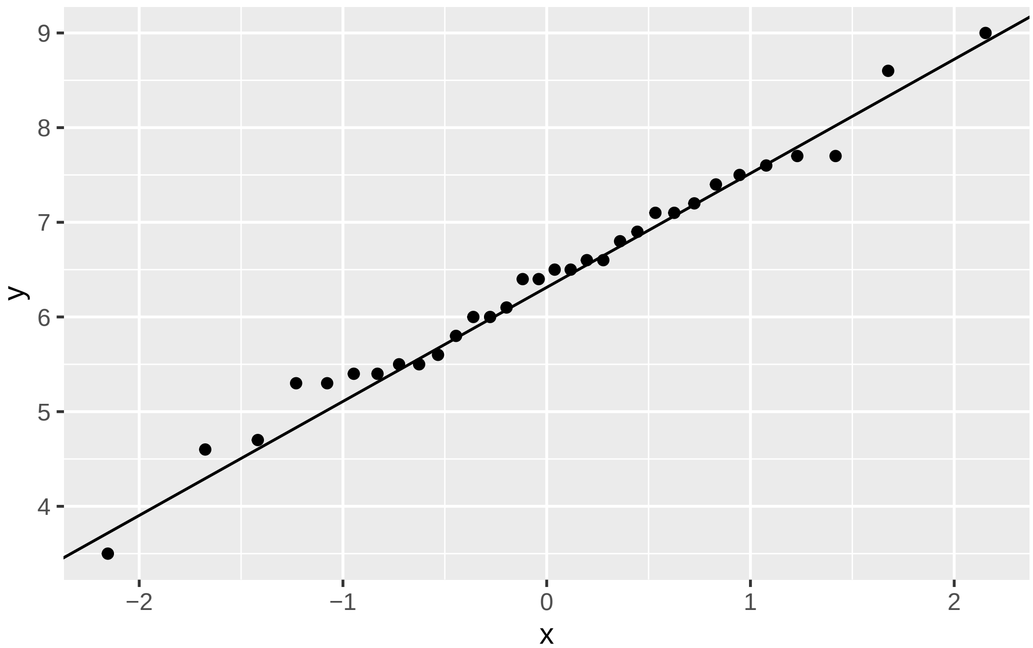
```r
# first layer: add scatter points
  geom_point(size = 2,
             stroke = 0.5,
             fill = "dodgerblue3",
             shape = 21)+

# relabel axis with units
  labs(x = "Ocean Temperature (°C)",
       y = "Giant Kelp Frond Growth Rate (cm/day)")+

# change theme
  theme_light()+
  theme(panel.grid = element_blank())
```



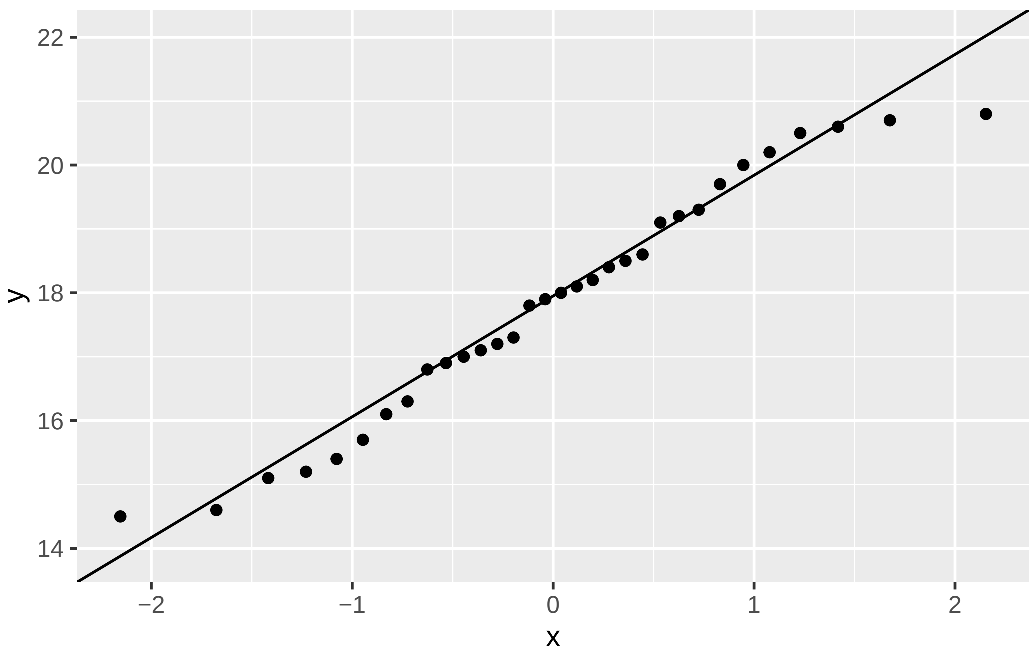
c. Check Assumptions and Run Test

```r
# check normality of kelp elongation rate
# base layer ggplot
ggplot(data = temp_kelp, mapping = aes(sample = kelp_elong)) +
```

```
# add sample points
  geom_qq() +

# add reference line
  geom_qq_line()
```




```
# check normality of temperature variable
# base layer ggplot
ggplot(data = temp_kelp, mapping = aes(sample = temp_c)) +

# add sample points
  geom_qq() +

# add reference line
  geom_qq_line()
```



Assumption of Pearson's correlation

To check the assumptions of a Pearson correlation test I evaluated the linearity, continuous data, normality and independence of observations.

I checked linearity by looking at the scatter plot generated in 1b; if I imagine a line from the lowest temp/highest growth rate point to one of the high-temp/low-growth points the data would fall around that line.

Temperature and growth rate are both continuous variables (they can be measured to decimal places or fractions).

To assess normality I created two QQ plots. One for temperature and one for kelp elongation rate. Looking at both diagnostic QQ plots, the data appears normally distributed since the data follows the line fairly close only deviating strongly in the tails which is acceptable.

Finally, the assumption of independent variables is a potential limitation since the study design does not explicitly say if the kelp fronds were isolated in separate, independently regulated tanks; however, we are going to proceed with the analysis.

Run a correlation test

```r
# Run Pearson Correlation test
cor.test(temp_kelp$kelp_elong, # predicted/y value
         temp_kelp$temp_c,     # predictor/x value
         method = "pearson")   # use pearson method
```

```
	Pearson's product-moment correlation

data:  temp_kelp$kelp_elong and temp_kelp$temp_c
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
       cor
-0.6877489
```

d. Results Communication

To evaluate the strength of the relationship between ocean temperature and giant kelp frond elongation rate, I used a Pearson's Correlation test. The data appears to be normally distributed based on the QQ plot diagnostic test meaning this parametric test is fitting.

We found a moderate negative relationship between giant kelp frond elongation rate (in cm/day) and ocean temperature (in degrees Celsius) (Pearson's r = -0.69, $t(30) = -5.19$, $p < 0.01$, $\alpha = 0.05$).

e. Test Implications

Our test suggest a moderate negative relationship between ocean temperature and kelp growth, meaning that giant kelp frond elongations rates decrease as water temperatures rise. As our ocean temperatures rise from climate change kelp will continue to have its growth stunted. To maximize giant kelp conservation efforts I would suggest prioritizing protecting the cooler areas of the ocean where giant kelp has higher chances of survival.

f. Run Other Correlation Test

```
# Run Spearman Correlation test
cor.test(temp_kelp$kelp_elong, # predicted/y value
         temp_kelp$temp_c,     # predictor/x value
         method = "spearman")  # change method to spearman
```

```
    Spearman's rank correlation rho

data:  temp_kelp$kelp_elong and temp_kelp$temp_c
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.6891684
```

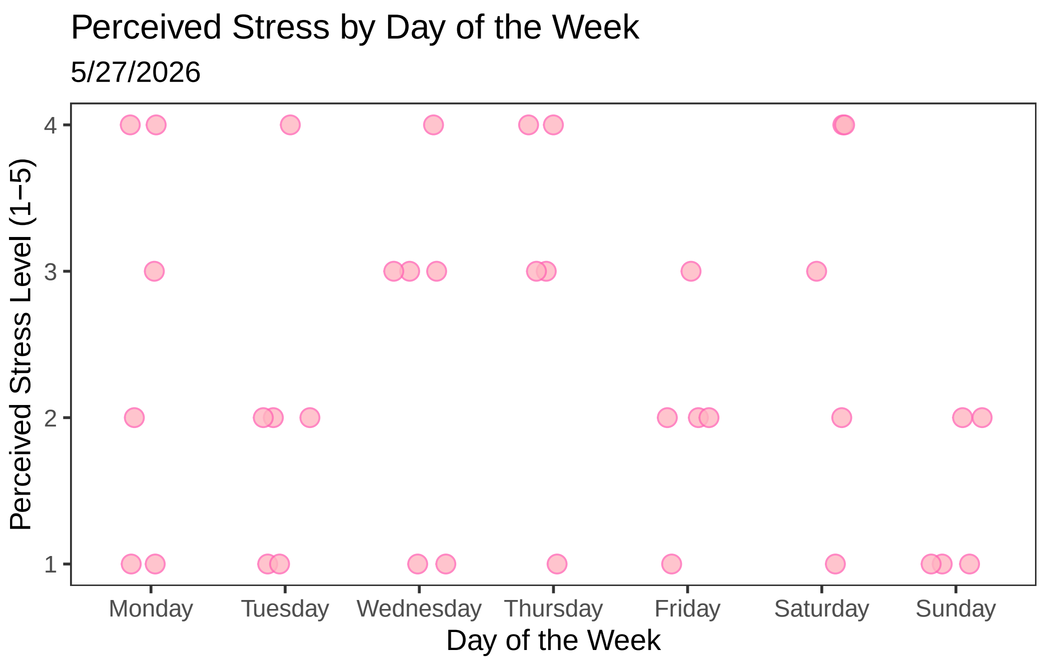
Both of these test lead me to reject the null hypothesis because they both yielded a p-value $<$ 0.01. Both methods lead to me to say there is a moderate negative relationship between ocean water temperature and giant kelp frond elongation rate because the Spearman rank coefficient (ρ = -0.689) is close to the Pearson correlation coefficient (Pearson's r = -0.688). This directly relates to the variables by confirming that as the continuous variable of ocean temperature increases, the giant kelp frond elongation rate decreases, a trend that remains robust whether measured linearly (with Pearson) or monotonically through ranked data (with Spearman).

Problem 2. Personal Data

a. Updating Visualizations

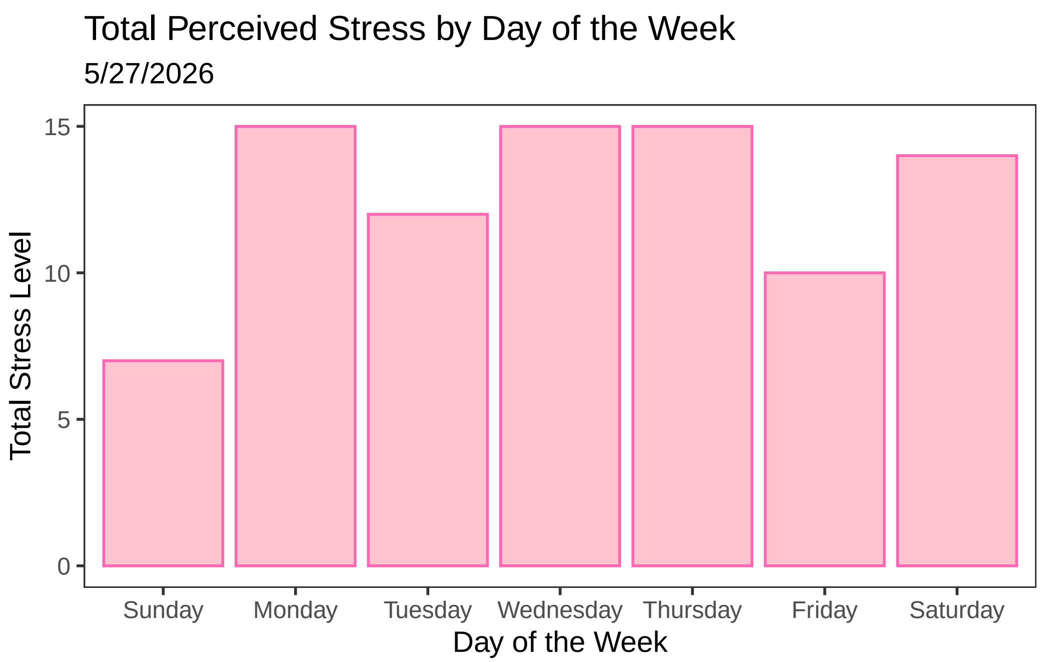
```
# organize day of week column so variable shows up in chronological order
# instead of alphabetical
my_data$day_of_the_week <- factor(my_data$day_of_the_week,
levels = c("Monday", "Tuesday", "Wednesday",
"Thursday", "Friday", "Saturday",
"Sunday"))

# create plot, assign x and y values.
# X is a categorical predictor variable, Y is response variable.
ggplot(my_data, aes(x = day_of_the_week, y = perceived_stress_level))+

# create jitter plot with custom color
```

```r
geom_jitter(height = 0,
            width = 0.2,
            shape = 21,
            size = 3,,
            fill = "lightpink",
            alpha = 0.8,
            color = "hotpink") +

# change graph title and axis titles
labs(title = "Perceived Stress by Day of the Week",
     subtitle = "5/27/2026",
x = "Day of the Week",
y = "Perceived Stress Level (1-5)")+

# change theme
  theme_bw()+

# remove grid
  theme(panel.grid = element_blank())
```



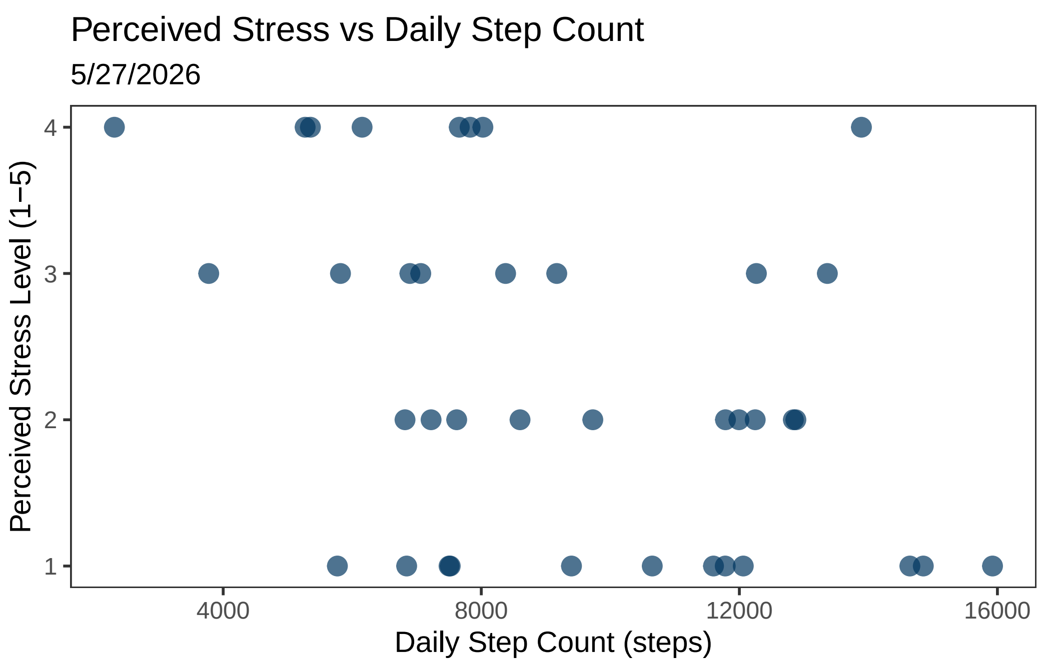
More interesting interpretation:

```
# keep your original factor sorting so days stay chronological
my_data$day_of_the_week <- factor(my_data$day_of_the_week, levels = c("Sunday", "Monday", "Tu

# map day to x, and stress to y
ggplot(my_data, aes(x = day_of_the_week, y = perceived_stress_level)) +

  # stat="summary" and fun="sum" will add up the stress levels automatically
  geom_bar(stat = "summary", fun = "sum", fill = "lightpink", color = "hotpink", alpha = 0.8)

  # update labels
  labs(title = "Total Perceived Stress by Day of the Week",
       subtitle = "5/27/2026",
       x = "Day of the Week",
       y = "Total Stress Level") +
  theme_bw() +
  theme(panel.grid = element_blank())
```




```
# create plot, assign x and y values
# X is my discrete predictor variable, Y is response variable.
ggplot(my_data, aes(x = daily_step_count, y = perceived_stress_level))+

# create plot with custom color
```

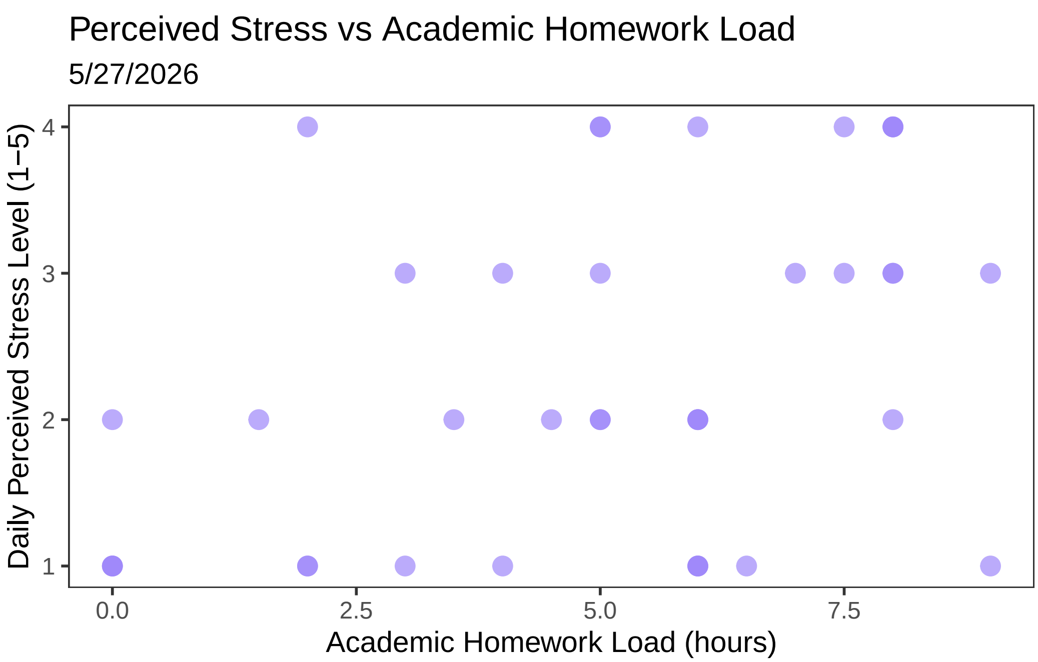
```r
geom_point(shape = 19,
            size = 3,
           alpha = 0.7,
           color = "#003660") +

# change graph title and axis titles
labs(title = "Perceived Stress vs Daily Step Count",
     subtitle = "5/27/2026",
x = "Daily Step Count (steps)",
y = "Perceived Stress Level (1-5)")+

# choosing non-default theme
  theme_bw()+

# remove grid
  theme(panel.grid = element_blank())
```



Other variables:

```r
# create plot, assign x and y values
# X is my continuous predictor variable, Y is response variable.
ggplot(my_data, aes(x = academic_homework_load, y = perceived_stress_level))+
```

```
# create plot with custom color
geom_point(shape = 19,
            size = 3,
           alpha = 0.7,
           color = "#9E86FA") +

# change graph title and axis titles
labs(title = "Perceived Stress vs Academic Homework Load",
     subtitle = "5/27/2026",
x = "Academic Homework Load (hours)",
y = "Daily Perceived Stress Level (1-5)")+

# choosing non-default theme
  theme_bw()+

# remove grid
  theme(panel.grid = element_blank())
```



b. Captions

Figure 1. Day of the week appears to have little impact on perceived stress level. Each pink circle represents a single day's self-reported stress level on a scale of 1-5. The points

are jittered and semi-transparent to show overlapping points. Days of the week are represented on the x axis, ordered from Monday to Sunday. There appears to be examples of low stress everyday of the week and high stress almost everyday. Tuesdays, Fridays, and Sundays tend to be lower stress. Wednesdays, Thursdays, and Saturdays appear to be the highest stress day of the week. Day of the week is the categorical predictor variable and stress is the categorical response variable.

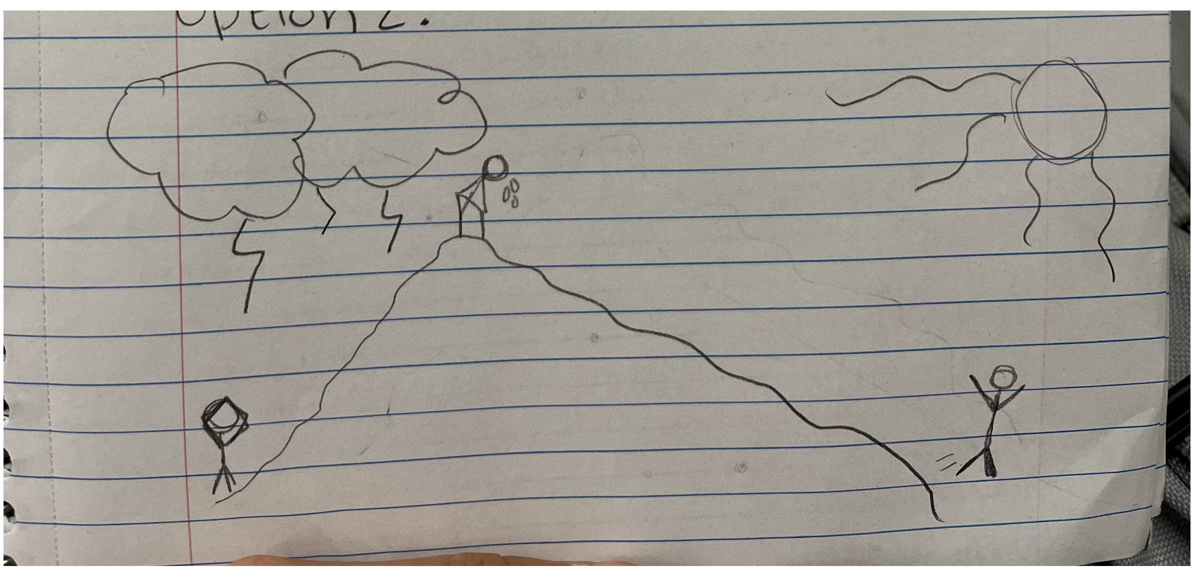
Figure 2. There appears to be a potential negative linear relationship between perceived stress level and daily step count. The navy blue points represent a single day's self-reported stress level on a scale of 1-5 and are ordered on the x axis by daily step count recorded by a fitbit watch. Daily step count is the discrete predictor variable and stress is the categorical response variable.

Problem 3. Affective Visualization

a. Affective visualization ideas

I envision an affective data visualization tha translates my daily step count and stress levels into a watercolor landscape. I was very inspired by the nature-focuesd work of Jill Pelto, but I want to give it a slightly cartoonish aesthetic. A central mountain could serve as the focal point symbolizing the physical steps it takes to climb to a calmer state of mind. I could use colors of the sky to convey emotion, for example using a sunny bright yellow to represent less stressed and gray-blue skies to represent more stressed.

b. Affective Sketch



c. Affective draft

d. Artist Statement

This watercolor painting represents the relationship between my daily step counts and my daily perceived stress level. I was inspired by Jill Pelto's paintings. I started my process by drafting with pencil and paper, created a digital rendition on my iPad to get a more solidified idea of color placement and finally, I mapped my design onto watercolor paper and painted my final draft.

e. Affective Visualization Presentation

View Presentation

Problem 4. Statistical Critique

a. Revisit and Summarize

The main research question is "What are the impacts of farming system (organic vs conventional) and landscape heterogeneity (large scale vs small scale) on butterfly diversity and species composition?" A Pearson correlation test is used in this paper. This test led the researchers to suggest that landscape heterogeneity is an important factor in butterfly diversity. A "significant" result would lead them to suggest there is a positive correlation between butterfly diversity and small-scale landscape heterogeneity.

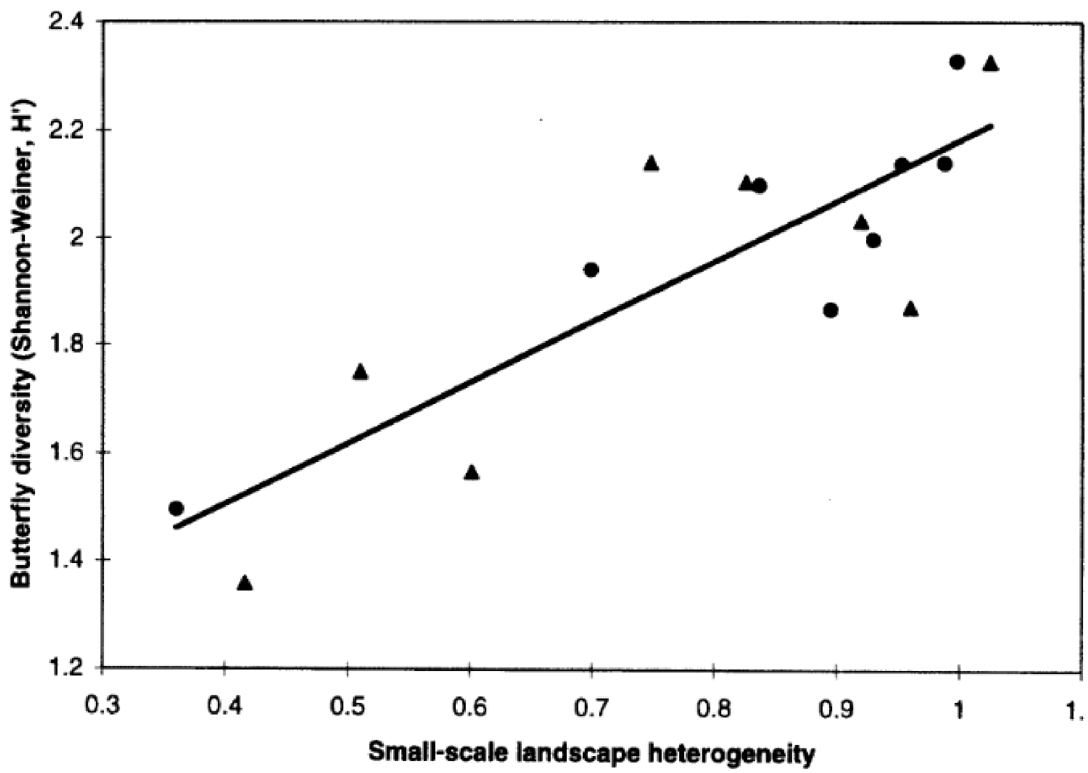


Fig. 2. The effect of small-scale landscape heterogeneity on butterfly species diversity (H′). Organic farms are shown as circles and conventional farms as triangles. (n = 16).

b. Visual Clarity

The authors visually represented their statistics in figures well. The one shown above clearly shows the positive linear relationship suggested by the pearson correlation test. The axes make sense; the predictor variable (landscape heterogeneity) is on the x axis and the response variable (butterfly diversity) on the y-axis. Summary statistics are not shown, but the model prediction is represented by a line and underlying data is shown.

c. Aesthetic Clarity

The authors kept visual clutter low and kept a high data to ink ratio by removing grid lines, keeping the background white, differentiating points by shape rather than color, only using a single trend line (no ribbon), only used axis lines (no box around the graph), and used concise but descriptive axis labels.

d. Recomendations

While it is not uncommon to keep the y-axis text sideways, it would improve instant readability to rotate it clockwise 90 degrees so the text is horizontal. Honestly I think the scale, shapes, size of points, and line width I would keep the same. If I could add color I may change the point colors to stand out from the line better, for example red triangles and blue circles. This would improve visual clarity particularly for the point where the one circle touches the line.